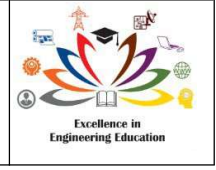




Pune Vidyarthi Griha's
COLLEGE OF ENGINEERING, TECHNOLOGY
AND MANAGEMENT, PUNE-09



Pune Vidyarthi Griha's College of Engineering, Technology and Management

Department of Artificial Intelligence and Data Science

Subject Name & Code:
417532(B): Business Intelligence

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Batch : B3

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ASSIGNMENT 4

AIM :

To perform data analysis and create insightful visualizations using advanced Excel features such as Pivot Tables, Charts, Conditional Formatting, thereby supporting decision-making in a business context.

PROBLEM STATEMENT :

A retail company collects monthly sales data across different regions and product categories. However, raw data alone does not provide meaningful insights. The company needs a way to analyze and visualize this data to identify sales trends, high-performing products, and regional performance. The goal is to help the management make informed decisions for inventory planning and marketing strategies.

THEORY :

Business Intelligence (BI) is a technology-driven process that helps organizations analyze data and present actionable information to aid in decision-making. One of the most accessible BI tools is **Microsoft Excel**, which, when used effectively, can transform raw data into meaningful insights.

Advanced Excel encompasses a suite of tools and features beyond basic spreadsheet functionality. These tools enable data analysts to perform complex calculations, manipulate large datasets, visualize data trends, and create interactive dashboards. Below are the key Advanced Excel concepts relevant to Business Intelligence:

1. **Pivot Tables:**

Pivot Tables are powerful tools used to automatically sort, count, and total data stored in one table or spreadsheet and create a second table displaying the summarized data. With just a few clicks, they can reveal patterns and relationships that are not immediately obvious.

2. **Pivot Charts:**

These are visual representations of Pivot Tables. They dynamically reflect changes in the Pivot Table and provide a graphical interface to explore patterns in the data.

3. **Conditional Formatting:**

This feature highlights cells based on specific criteria (e.g., color scales,

data bars, or custom rules). It visually emphasizes high or low values, trends, and outliers, making it easier to interpret data at a glance.

4. Data Validation:

Used to create dropdown lists or limit the type of data that can be entered in cells. This helps in standardizing data entry and preventing errors.

5. Charts (Bar, Line, Pie, etc.):

Excel supports various chart types that are useful for comparing data (bar), showing trends (line), or illustrating proportions (pie).

6. Lookup Functions (VLOOKUP, HLOOKUP, INDEX-MATCH):

These functions fetch data from different parts of a dataset or other sheets, allowing for flexible data integration and comparison.

7. Dashboards:

A dashboard in Excel combines multiple charts, tables, KPIs, and filters into a single interactive sheet. It serves as a snapshot for decision-makers to assess performance at a glance.

By using these features together, users can derive insights from data and present them in a way that's easy to interpret and act upon.

PROCEDURE

Step 1: Importing the Data

- Open Microsoft Excel and import the dataset (e.g., monthly sales data).
- Ensure that the data is structured in tabular form with clearly labeled columns like Region, Product Category, Month, Units Sold, Sales, and Profit.

Step 2: Data Cleaning and Preparation

- Remove any duplicate records or empty rows.
- Standardize column formats (e.g., currency for Sales and Profit, text for Region).
- Use Excel's **Format as Table** option to apply structured formatting.

Step 3: Data Validation

- Apply dropdown menus for columns like Region and Product Category using **Data Validation > List**.
This ensures consistency in data entry.

Step 4: Applying Conditional Formatting

- Highlight key metrics using color scales or rules:
 - Green for high profits
 - Red for very low sales
- Use **Data Bars** to show profit margins visually.

Step 5: Creating Pivot Tables

- Insert Pivot Tables to analyze:
 - Sales by Region
 - Monthly Profit Trends
 - Product Category performance
- Place the Pivot Tables on separate sheets for clarity.

Step 6: Visualizing the Data

- Create charts based on Pivot Table outputs:
 - **Bar Chart** for Regional Sales
 - **Line Chart** for Monthly Trends
 - **Pie Chart** for Category-wise Sales
- Customize with titles, data labels, and colors.

Step 7: Building the Dashboard

- Create a new sheet titled “Dashboard.”
- Arrange charts neatly with appropriate spacing.
- Optionally add slicers (if using Excel 2013 or later) to allow users to filter the dashboard dynamically by Region or Month.

Step 8: Drawing Insights

- Interpret the visualizations:
 - Identify which region performs best
 - Spot seasonal trends in monthly sales
 - See which product category generates the most revenue
- Note these observations for use in the final report or presentation.

Step 9: Saving and Exporting

- Save the Excel workbook with all sheets: raw data, Pivot summaries, charts, and dashboard.
- Optionally export the dashboard as a PDF or image for sharing.

CONCLUSION

Using Advanced Excel for data analysis and visualization enables quick, accurate, and meaningful interpretation of business data. In this assignment, we used various Excel tools to transform raw sales data into actionable insights. The created dashboard and visualizations supported decision-making processes such as identifying top-performing regions and optimizing product strategies. Thus, Advanced Excel proves to be a powerful Business Intelligence tool in today's data-driven business environment.

Date		Region	Salesperson	Product	Units Sold	Unit Price	Revenue	Performance
2024-03-0	East	Alice	Monitor	3	60000	180000	Normal	
2024-02-2	East	Alice	Laptop	1	20000	20000	Normal	
2024-01-2	West	Diana	Laptop	15	30000	450000	High	
2024-01-0	West	Alice	Tablet	13	20000	260000	Normal	
2024-03-2	West	Bob	Tablet	13	30000	390000	Normal	
2024-02-0	South	Diana	Tablet	2	20000	40000	Normal	
2024-01-2	North	Alice	Laptop	10	30000	300000	Normal	
2024-02-0	East	Bob	Phone	4	20000	80000	Normal	
2024-02-1	South	Diana	Laptop	2	20000	40000	Normal	
2024-03-0	West	Charlie	Monitor	5	30000	150000	Normal	
2024-03-1	South	Alice	Tablet	13	25000	325000	Normal	
2024-01-2	West	Alice	Monitor	1	20000	20000	Normal	
2024-01-1	East	Alice	Laptop	12	30000	360000	Normal	
2024-02-0	North	Charlie	Tablet	9	30000	270000	Normal	
2024-02-0	South	Alice	Tablet	14	20000	280000	Normal	
2024-02-2	East	Alice	Monitor	12	20000	240000	Normal	
2024-03-3	West	Charlie	Phone	8	60000	480000	High	
2024-02-1	North	Charlie	Tablet	7	30000	210000	Normal	
2024-02-2	South	Alice	Tablet	4	60000	240000	Normal	
2024-03-1	East	Diana	Tablet	8	60000	480000	High	
2024-02-2	South	Bob	Tablet	15	20000	300000	Normal	
2024-02-1	East	Diana	Tablet	3	30000	90000	Normal	
2024-01-0	West	Bob	Monitor	10	60000	600000	High	
2024-02-0	South	Alice	Tablet	7	30000	210000	Normal	
2024-03-0	South	Charlie	Monitor	9	30000	270000	Normal	
2024-03-0	West	Charlie	Laptop	5	60000	300000	Normal	
2024-03-1	West	Charlie	Tablet	2	25000	50000	Normal	
2024-02-1	East	Diana	Tablet	6	20000	120000	Normal	
2024-03-0	South	Diana	Monitor	8	25000	200000	Normal	
2024-01-1	South	Bob	Phone	7	20000	140000	Normal	

Total Revenue for east region: 7930000

